班级：________ 姓名：_________学号：_________

Thermodynamics

1．What is the pressure inside a 35.0-L container holding 105.0kg of argon (Ar) gas at 20.0^{0}C? (molar mass of Ar: 40g/mol) [1.8×10^3atm]

2．If 61.5L of oxygen (O_2) at 18.0^0 C and absolute pressure of 2.45atm are compressed to 48.8 L and at the same time the temperature is raised to 50.0^{0}C, what will the new pressure be? [$3.43atm\,(3.47\times10^5 Pa)$]

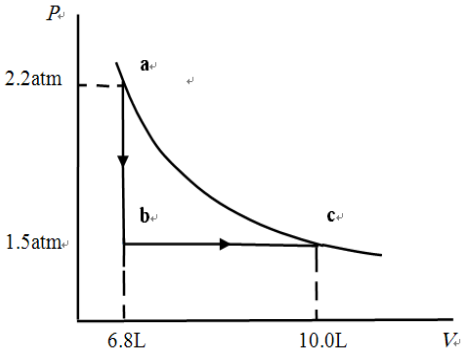
3．If the pressure in a gas is doubled while its volume is held constant, by what factor does v_{rms} (方均根速率) change? [$\sqrt{2}$]

4. **It is found that the most probable speed of molecules in a gas when it has (uniform) temperature T_2 is the same as the rms speed (方均根速率） of the molecules in this gas when it has (uniform) temperature T_1, Calculate T_2/T_1.**[3/2]

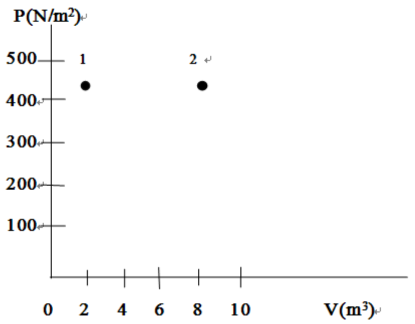
5. **What is the collision frequency for N_2 molecules in air at T=20.0^{0}C and** $P=1.0\times10^{-2}\text{atm}$ **?** $(r=1.5\times10^{-10}m)$ [$4.71\times10^{7}/s$]

6. **What is the rms speed of nitrogen (N_2) molecules contained in a 12.8m^3 at 3.42atm if the total amount of nitrogen is 1800 mol?** [$514m/s$]

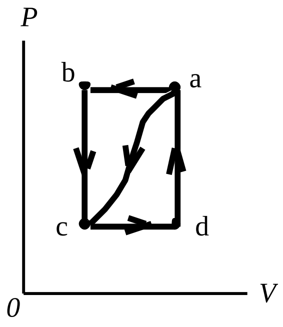
7. **Consider the following two-step process. Heat is allowed to flow out of an ideal gas at constant volume so that its pressure drops from 2.2 atm to 1.5 atm. Then the gas expands at constant pressure, from a volume of 6.8L to 10.0L, where the temperature reaches its original value as shown in figure. Calculate (a) the total work done by the gas in the process, (b)the change in internal energy of the gas in the process, and (c) the total heat flow into or out of the gas.** **[(a)** $4.9\times10^{2}J$ **(b) 0** **(c)** $4.9\times10^{2}J$ **]**



8. **The PV program in Fig. shows two possible states of a system containing 1.45 moles of a monatomic deal gas. (P_1=P_2=450N/m^2,V_1=2.00m^3,V_2=8.00m^3). (a)Draw the process which depicts an isobaric expansion from state 1 to state 2 and the label this process (A). (b)Find the work done by the gas and the change in internal energy of the gas in process (A). (c)Draw the process which depicts an isothermal expansion from state 1 to the volume V_2 followed by an isochoric increase in temperature to state 2 and label this two-step process(B).(d)Find the change in internal energy of the gas for the two-step process(B).[(a)(c) 略 (b) 2700J,** $4.05\times10^{3}J$ **(d)** $4.05\times10^{3}J$ **]**



9. **When a gas is taken from a to c along the curved path in Fig., the work done by the gas is $W = -35$J and the heat added to the gas is $Q = -63$J. Along path abc, the work done is $W = -48$J. (a)What is Q for path abc? (b)If $P_c = \frac{1}{2} P_b$, what is W for path cda? (c)What is Q for path cda? (d)What is U_a-U_c? (e)If $U_d - U_c = 5$J, what is Q for path da?**
[(a) -76J (b) 24J (c) 52J (d)28J (e) 23J]



10. **A sample of 770mol of nitrogen gas is maintained at a constant pressure of 1.00 atm in a flexible container. The gas is heated from 40°C to180°C(a)the heat added to the gas.(b) the work done by the gas and (c) the change in internal energy.**
[(a) $3.14\times10^{6} J$ **(b)** $8.96\times10^{5} J$ **(c)** $2.24\times10^{6} J$ **]**

11. **Suppose 4.00 mol of an ideal diatomic gas, with molecular rotation but not oscillation, experienced a temperature increase of 60.0K under constant-pressure conditions. What are (a) the energy transferred as heat *Q*, (b) the change ΔU in internal energy of the gas, (c) the work *W* done by the gas, and (d) the change ΔK in the total translational kinetic energy of the gas ? [(a) 6.98×10^{3} J (b) 4.99×10^{3} J (c) 1.99×10^{3} J (d) 2.99×10^{3} J.]**

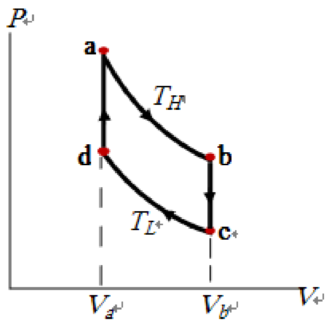
12. **An ideal gas is suddenly allowed to expand freely so that the ratio of its new volume V_1 to its initial volume V_0 is V_1/V_0=5.00. The gas is then adiabatically(绝热的) compressed back to its initial volume V_0, leaving it with a pressure p_2 that is $(5.00)^{0.40}$ times its initial pressure p_0. (a) Is the gas monatomic, diatomic or polyatomic? What is the ratio of the final value of the average kinetic energy per molecule to the initial value (b) after the free expansion and (c) after the adiabatic compression?**
[(a) diatomic (b) 1 (c) 1.9]

13. The temperature of 3.00mol of an ideal diatomic gas is increased by 40.0℃ without the pressure of the gas changing. The molecules in the gas rotate but do not oscillate. (a)How much energy is transferred to the gas as heat? (b) What is the change in the internal energy of the gas? (c) How much work is done by the gas? (d) By how much does the translational kinetic energy of the gas increase?

[(a) 3.49×10^{3} J. (b) 2.49×10^{3} J. (c) 997J (d) 1.49×10^{3} J.]

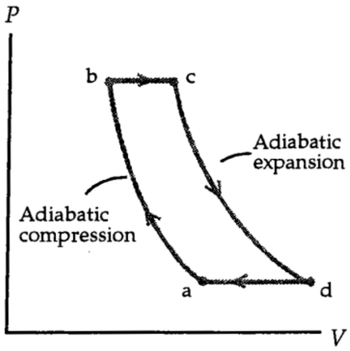
14. The Stirling cycle, shown in Fig., is useful to describe external combustion engines as well as solar-power systems. Find the efficiency of the cycle in terms of the parameters shown, assuming a monatomic gas as the working substance（工作物质）. The processes ab and cd are isothermal whereas bc and da are constant volume. How does it compare to the Carnot efficiency?

[$e = 1 - \dfrac{\frac{3}{2}(T_H - T_L) + T_L \ln\frac{V_b}{V_a}}{\frac{3}{2}(T_H - T_L) + T_H \ln\frac{V_b}{V_a}} < 1 - \dfrac{T_L}{T_H} (e_{carnot})$]



15. A gas turbine operates under the Brayton cycle, which is depicted in the PV diagram of Fig. In process *ab* the air-fuel mixture undergoes an adiabatic compression. This is followed, in process *bc*, with an isobaric (constant pressure) heating, by combustion. Process cd is an adiabatic expansion with expulsion of products to atmosphere. The return step, da, takes place at constant pressure. If the working gas behaves like an ideal gas, show that the efficiency of Brayton Cycle is

$$e = 1 - \left(\frac{P_b}{P_a}\right)^{\frac{1-\gamma}{\gamma}}$$



16. An ideal gas (1.0 mol) is the working substance in an engine that operates on the cycle shown in Fig.. Processes BC and DA are reversible and adiabatic.(a) Is the gas monatomic, diatomic, or polyatomic? (b)What is the engine efficiency?

[(a)monatomic (b) 75%]